

SAUTRIK ROY

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EDUCATION

SRM Institute of Science and Technology (SRM IST)

Kattankulathur, Chennai, India

B.Tech in Computer Science & Engineering (Core)

Expected May 2029 (Batch 2025–2029)

- **CGPA:** 8.77 / 10.0 | **Coursework:** Data Structures & Algorithms, Object-Oriented Programming (C++), Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering.

TECHNICAL SKILLS

Programming Languages: C/C++, TypeScript, JavaScript (ES6+), Python, Java, SQL

Frontend & Web: React 18, Next.js, TanStack Start, HTML5, CSS3, Tailwind CSS, WebRTC, PWA, Context API

Backend & Cloud: Node.js, Express.js, REST APIs, WebSockets, Supabase, Microsoft Azure (AZ-900), Microservices

Databases & Storage: PostgreSQL, MySQL, MongoDB, IndexedDB, Database Management Systems (DBMS), Query Optimization

DevOps, Testing & Tools: Git/GitHub, GitHub Actions, CI/CD Pipelines, Docker, Linux, macOS, Postman, Vite, Vitest/Jest

CS Fundamentals: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, System Design, Agile/Scrum

EXPERIENCE & OPEN SOURCE

Microsoft Learn Student Ambassador (MLSA) — Campus Ambassador

Jan 2025 – Present

- Spearheaded 4+ technical workshops empowering 200+ campus developers in Microsoft Azure cloud architecture, Docker containerization fundamentals, and full-stack software engineering standards.
- Mentored 50+ junior students in developer environments, promoting collaborative Git/GitHub CI/CD workflows, peer code reviews, and modern system design practices.

GirlScript Summer of Code (GSSoC) — Open Source Contributor

2025

- Merged 8+ production pull requests across distributed open-source repositories; resolved 5 frontend bugs and optimized component render cycles in React and TypeScript codebases.
- Collaborated with repository maintainers via Git/GitHub in an Agile setting, verifying PRs through GitHub Actions CI/CD workflows and authoring unit tests using Vitest/Jest.

TECHNICAL PROJECTS

Sentinel — AI Armed Forces Wellbeing & Telemedicine Platform (SIH)

React 18 • Node.js • Express.js • Supabase • WebRTC • PWA

► [Live](#) | [GitHub](#)

- Architected an AI-powered cognitive readiness platform for India's Armed Forces (SIH Ministry of Home Affairs); engineered predictive scoring algorithms tracking 5 longitudinal signals (stress, sleep, fatigue, mood, energy) to screen for burnout.
- Built Node.js/Express.js REST APIs with Supabase backend, encrypted WebRTC video telehealth, and an offline-first PWA caching records in IndexedDB—preserving 100% of offline assessments with sub-second local response time.

Loop Beyond Limits — Full-Stack Music Streaming Platform

► [Live](#) | [GitHub](#)

TypeScript • React • TanStack Start • Supabase • WebSockets

- Architected an ad-free music streaming platform utilizing React, TanStack Start, and Supabase; implemented Server-Side Rendering (SSR) to cut initial page load time by ~40% by eliminating client-side render cycles.
- Engineered secure authentication flows, user validation, and Row-Level Security (RLS) policies with real-time WebSocket state synchronization in PostgreSQL/Supabase to enable concurrent multi-user access without data conflicts.

Quizzify — AI-Powered Competitive Assessment Platform

► [Live](#) | [GitHub](#)

C++ • Google Gemini API • REST APIs • OOP

- Developed a high-speed C++ parsing backend to extract text from multi-page PDFs, integrating Google Gemini API via REST endpoints to automatically generate structured quiz questions in under 10 seconds.
- Applied OOP inheritance and polymorphism in C++ to implement modular Test Mode and Practice Mode architectures with shared base class logic, eliminating code duplication and streamlining mode extensibility.

LegacyLens 2.0 — AI Natural Language to Database Query Interface

► [Live](#) | [GitHub](#)

TypeScript • MySQL • MongoDB • NLP

- Built a web interface converting plain-English input into valid MySQL and MongoDB queries using NLP techniques, allowing users to query relational and document databases without writing SQL.
- Implemented parameterized queries and input validation for SQL injection prevention; benchmarked query generation against 10+ query patterns to verify accuracy across both relational and non-relational databases.

HONORS & CERTIFICATIONS

• **Smart India Hackathon (SIH) 2026:** Shortlisted among Top 100 Teams (out of 130 teams / 780 competitors) in SRM University Internal Hackathon for Project SENTINEL (Defense AI Wellbeing System) ► [Live Demo](#).

• **Certifications:** Microsoft Certified: Azure Fundamentals (AZ-900) • Prompt Engineering with GitHub Copilot • Intro to Gen AI (Google).

• **Languages:** English (Professional), Bengali (Native), Hindi (Conversational), German (A1).